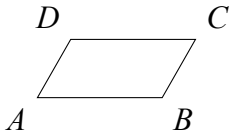
1 Vectors I

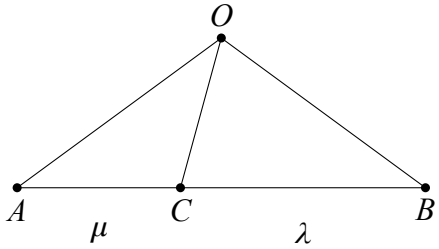
DRAFT: Working draft. All examples are taken from your Vectors Summary Notes; the workings are mine, checked with sympy.

1.1 Basic Formulas

	Formula	Note
Magnitude	$\lvert\mathbf{a}\rvert = \sqrt{x^2+y^2+z^2}$	
Unit vector	$\hat{\mathbf{a}} = \dfrac{\mathbf{a}}{\lvert\mathbf{a}\rvert}$	$\lvert\hat{\mathbf{a}}\rvert = 1$
Parallel vectors	$\mathbf{a} = k\mathbf{b}$ $\quad$ $\mathbf{a}\times\mathbf{b} = \mathbf{0}$	k scalar constant
Collinear A, B, C	$\overrightarrow{AB} \parallel \overrightarrow{BC}$ (or any two of $\overrightarrow{AB}, \overrightarrow{BC}, \overrightarrow{AC}$)	Try drawing the diagram and take any two vectors, to show that they are parallel.
Parallelogram $ABCD$	$\overrightarrow{AB} = \overrightarrow{DC}$, $\quad$ $\overrightarrow{AD} = \overrightarrow{BC}$	

The Ratio Theorem

Let the points A, B and C have position vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ respectively, and suppose that $\lambda AC = \mu CB$. Then the position vector of C is given by the formula

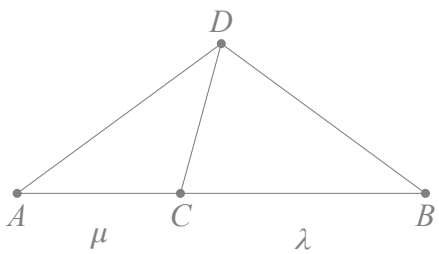


Ratio Theorem

$$\mathbf{c} = \frac{\lambda\mathbf{a} + \mu\mathbf{b}}{\lambda + \mu}$$

$AC : CB = \mu : \lambda$, so C divides AB with μ beside A and λ beside B.

The same theorem, starting from a point D instead of O:

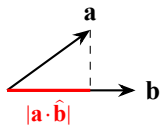
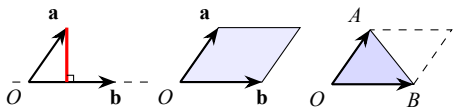


Ratio Theorem

$$\overrightarrow{DC} = \frac{\lambda\overrightarrow{DA} + \mu\overrightarrow{DB}}{\lambda + \mu}$$

1.2 Scalar Product and Vector Product

$$\mathbf{a}=a_1\mathbf{i}+a_2\mathbf{j}+a_3\mathbf{k}, \qquad \mathbf{b}=b_1\mathbf{i}+b_2\mathbf{j}+b_3\mathbf{k}, \qquad \theta=\text{non-reflex } \angle AOB$$

	Scalar Product (Dot)	Vector Product (Cross)
Definition	$\mathbf{a}\cdot\mathbf{b}=a_1b_1+a_2b_2+a_3b_3$	$\mathbf{a}\times\mathbf{b}=\begin{pmatrix} a_2b_3-a_3b_2 \\ -(a_1b_3-a_3b_1) \\ a_1b_2-a_2b_1 \end{pmatrix}$
Result	scalar	vector, $\perp$ to both $\mathbf{a}$ and $\mathbf{b}$
Formula	$\mathbf{a}\cdot\mathbf{b}=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\theta$	$\lvert\mathbf{a}\times\mathbf{b}\rvert=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\sin\theta$
Angle	$\cos\theta=\dfrac{\mathbf{a}\cdot\mathbf{b}}{\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert}$	$\sin\theta=\dfrac{\lvert\mathbf{a}\times\mathbf{b}\rvert}{\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert}$ this method only finds the basic angle possible, but usually undesirable
Properties	$\mathbf{a}\cdot\mathbf{b}=\mathbf{b}\cdot\mathbf{a}$ $\mathbf{a}\cdot(\mathbf{b}+\mathbf{c})=\mathbf{a}\cdot\mathbf{b}+\mathbf{a}\cdot\mathbf{c}$ $\lambda(\mathbf{a}\cdot\mathbf{b})=(\lambda\mathbf{a})\cdot\mathbf{b}=\mathbf{a}\cdot(\lambda\mathbf{b})$ $\mathbf{a}\cdot\mathbf{a}=\lvert\mathbf{a}\rvert^2$	$\mathbf{a}\times\mathbf{b}=-\mathbf{b}\times\mathbf{a}$ $\mathbf{a}\times(\mathbf{b}+\mathbf{c})=\mathbf{a}\times\mathbf{b}+\mathbf{a}\times\mathbf{c}$ $\lambda(\mathbf{a}\times\mathbf{b})=(\lambda\mathbf{a})\times\mathbf{b}=\mathbf{a}\times(\lambda\mathbf{b})$ $\mathbf{a}\times\mathbf{a}=\mathbf{0}$
Test	$\mathbf{a}\cdot\mathbf{b}=0 \iff \mathbf{a}\perp\mathbf{b}$ >0 acute, <0 obtuse, $=0$ right angle	$\mathbf{a}\times\mathbf{b}=\mathbf{0} \iff \mathbf{a}\parallel\mathbf{b}$ (nonzero $\mathbf{a}$, $\mathbf{b}$)
Applications	Length of projection $=\lvert\mathbf{a}\cdot\hat{\mathbf{b}}\rvert$ Projection vector $=(\mathbf{a}\cdot\hat{\mathbf{b}})\hat{\mathbf{b}}$	Perpendicular distance $=\lvert\mathbf{a}\times\hat{\mathbf{b}}\rvert$ Area of $\triangle OAB=\frac{1}{2}\lvert\mathbf{a}\times\mathbf{b}\rvert$ Area of parallelogram $=\lvert\mathbf{a}\times\mathbf{b}\rvert$
Geometrical meaning	 $\lvert\mathbf{a}\cdot\hat{\mathbf{b}}\rvert$ = length of projection of $\mathbf{a}$ onto $\mathbf{b}$	 perpendicular distance from A to line along $\mathbf{b}$ $\lvert\mathbf{a}\times\hat{\mathbf{b}}\rvert$ area of parallelogram, sides $\mathbf{a}$, $\mathbf{b}$ $\lvert\mathbf{a}\times\mathbf{b}\rvert$ 2× area of $\triangle OAB$ $\lvert\mathbf{a}\times\mathbf{b}\rvert$
Note	$\mathbf{a}\cdot\mathbf{b}\cdot\mathbf{c}$ meaningless; no cancellation law	no cancellation law $(\mathbf{a}\times\mathbf{b})\times\mathbf{c}$ is a vector, but triple cross products are usually not tested

Note

Projection of $\mathbf{a}$ onto $\mathbf{b}$ $\neq$ projection of $\mathbf{b}$ onto $\mathbf{a}$. Perpendicular distance from A to OB $\neq$ from B to OA. Triangles without O as a vertex: $\frac{1}{2}\lvert\overrightarrow{AB}\times\overrightarrow{AC}\rvert$.

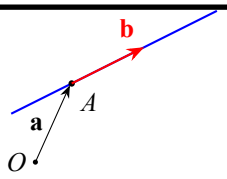
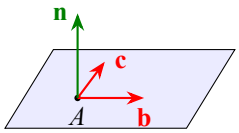
Useful identity

For any vectors $\mathbf{a}$, $\mathbf{b}$: $(\mathbf{a}\times\mathbf{b})\cdot\mathbf{a}=0$ and $(\mathbf{a}\times\mathbf{b})\cdot\mathbf{b}=0$

$\because \mathbf{a}\times\mathbf{b}$ is perpendicular to $\mathbf{a}$ and $\mathbf{b}$.

2 Vectors II

2.1 Equations of Lines and Planes

	Vector	Cartesian	What you need
Line	 $\mathbf{r}=\mathbf{a}+\lambda\mathbf{b},\ \lambda\in\mathbb{R}$ $(\mathbf{r}-\mathbf{a})\times\mathbf{b}=\mathbf{0}$ (same line: $\mathbf{r}-\mathbf{a}$ // $\mathbf{b}$)	$\frac{x-a}{d}=\frac{y-b}{e}=\frac{z-c}{f}$ $\mathbf{b}=\begin{pmatrix}d\\e\\f\end{pmatrix}$; if $d=0$: $x=a$ separately all coefficients of x, y and z in the numerators must be 1	**Need:** 1 point + 1 direction or 2 points A, B: $\mathbf{b}=\overrightarrow{AB}$
Plane	 $\mathbf{r}=\mathbf{a}+\lambda\mathbf{b}+\mu\mathbf{c}$ $\mathbf{r}\cdot\mathbf{n}=d$ (scalar product form)	$ax+by+cz=d$	**Need:** 1 point + 2 non-parallel directions // plane or 1 point + normal $\mathbf{n}$ $\mathbf{n}=\mathbf{b}\times\mathbf{c}$, $d=\mathbf{a}\cdot\mathbf{n}$ 3 points: $\mathbf{b}=\overrightarrow{AB}$, $\mathbf{c}=\overrightarrow{AC}$; contains a line: use its direction

$\mathbf{a}$: position vector of any point on the line/plane. $\mathbf{b}$, $\mathbf{c}$: any two linearly independent directions parallel to the plane.

The equation of any line is not unique.

Note

To check if a given point A lies on

- a line: $\mathbf{a}_A=\mathbf{a}+\lambda\mathbf{b}$ (one λ satisfies all 3 components)
- a plane: $\mathbf{a}_A\cdot\mathbf{n}=d$

Example 2.1

Line through $(1,2,3)$, $(3,2,2)$

$$\mathbf{b}=\begin{pmatrix}2\\0\\-1\end{pmatrix}$$

$$\mathbf{r}=\begin{pmatrix}1\\2\\3\end{pmatrix}+\lambda\begin{pmatrix}2\\0\\-1\end{pmatrix}$$

$$\frac{x-1}{2}=\frac{z-3}{-1},\quad y=2$$

Example 2.2

Plane through $B(0,1,1)$, $C(1,1,3)$, $D(2,0,2)$

$$\mathbf{n}=\overrightarrow{BC}\times\overrightarrow{BD}=\begin{pmatrix}1\\0\\2\end{pmatrix}\times\begin{pmatrix}2\\-1\\1\end{pmatrix}=\begin{pmatrix}2\\3\\-1\end{pmatrix}$$

$$d=\begin{pmatrix}0\\1\\1\end{pmatrix}\cdot\begin{pmatrix}2\\3\\-1\end{pmatrix}=2$$

$$2x+3y-z=2$$

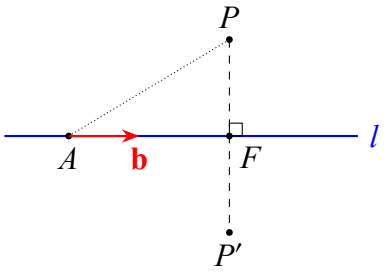
2.2 Interactions Map

	Point	Line	Plane
Point	distance $\lvert\overrightarrow{AB}\rvert$		
Line	• on line? §2.1 • foot F §2.3 • shortest distance §2.3 • reflection §2.3	• // / intersect / skew §2.4 • point of intersection §2.4 • angle §2.5 • reflect line in line §2.6	
Plane	• on plane? §2.1 • foot F §2.3 • shortest distance §2.3 • reflection §2.3	• // / lies in / intersect §2.4 • point of intersection §2.4 • angle §2.5 • reflect line in plane §2.6	• // / same / intersect §2.4 • line of intersection §2.4 • angle §2.5 • distance (//) §2.7

Always draw the diagram first.

2.3 Foot / Distance / Reflection

Point P – Line $\mathbf{r}=\mathbf{a}+\lambda\mathbf{b}$

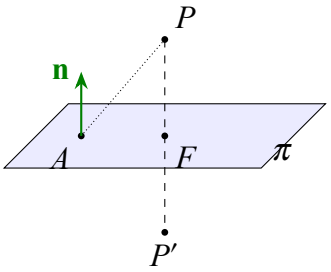


1. $\overrightarrow{OF}=\mathbf{a}+\lambda\mathbf{b}$ for some $\lambda\in\mathbb{R}$
2. $\overrightarrow{PF}\cdot\mathbf{b}=0$ ⇒ solve for λ
3. $\overrightarrow{OF}=\dfrac{\overrightarrow{OP}+\overrightarrow{OP'}}{2}$ (midpoint / ratio theorem)

Distance

$$PF=\lvert\overrightarrow{AP}\times\hat{\mathbf{b}}\rvert$$

Point P – Plane $\mathbf{r}\cdot\mathbf{n}=d$



1. l_{PF}: $\mathbf{r}=\mathbf{p}+\lambda\mathbf{n}$, $\lambda\in\mathbb{R}$
2. $(\mathbf{p}+\lambda\mathbf{n})\cdot\mathbf{n}=d$ (intersection between line & plane)
   ⇒ solve for λ
3. $\overrightarrow{OF}=\dfrac{\overrightarrow{OP}+\overrightarrow{OP'}}{2}$ (midpoint / ratio theorem)

Distance

$$PF=\lvert\overrightarrow{AP}\cdot\hat{\mathbf{n}}\rvert$$

Reflection (and foot of perpendicular) of point to line

Example 2.3

Find the foot of the perpendicular N from $A(3, -2, -1)$ to the line l_1, and the reflection A' of A in l_1, where

$$l_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 0 \\ -1 \end{pmatrix}$$

$$\overrightarrow{AN} = \begin{pmatrix} -2+3\lambda \\ 4 \\ 4-\lambda \end{pmatrix}$$

$$\overrightarrow{AN} \cdot \begin{pmatrix} 3 \\ 0 \\ -1 \end{pmatrix} = 0$$

$$10\lambda - 10 = 0 \Rightarrow \lambda = 1$$

$$\therefore N = (4, 2, 2)$$

$$\overrightarrow{OA'} = 2\begin{pmatrix} 4 \\ 2 \\ 2 \end{pmatrix} - \begin{pmatrix} 3 \\ -2 \\ -1 \end{pmatrix}$$

$$\therefore A' = (5, 6, 5)$$

Reflection (and foot of perpendicular) of point to plane

Example 2.4

Find the foot of the perpendicular F from $P(7, 0, 3)$ to the plane p, and the reflection P' of P in p, where

$$p : \mathbf{r} \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = 1$$

$$\left(\begin{pmatrix} 7 \\ 0 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} \right) \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = 1$$

$$13 + 6\lambda = 1 \Rightarrow \lambda = -2$$

$$\therefore F = (5, -2, -1)$$

$$\overrightarrow{OP'} = 2\begin{pmatrix} 5 \\ -2 \\ -1 \end{pmatrix} - \begin{pmatrix} 7 \\ 0 \\ 3 \end{pmatrix}$$

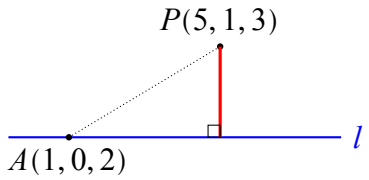
$$\therefore P' = (3, -4, -5)$$

Shortest distance from point to line

Example 2.5

Find the shortest distance from $P(5,1,3)$ to the line

$$l: \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$$



$$\text{shortest distance} = |\overrightarrow{AP} \times \hat{\mathbf{b}}|$$

$$= \left| \begin{pmatrix} 4 \\ 1 \\ 1 \end{pmatrix} \times \frac{1}{3} \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} \right|$$

$$= \left| \frac{1}{3} \begin{pmatrix} 3 \\ -6 \\ -6 \end{pmatrix} \right| = \left| \begin{pmatrix} 1 \\ -2 \\ -2 \end{pmatrix} \right|$$

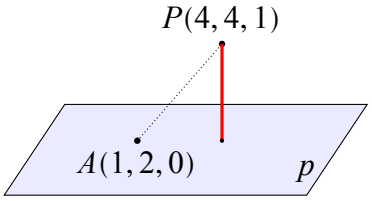
$$= 3$$

Shortest distance from point to plane

Example 2.6

Find the shortest distance from $P(4,4,1)$ to the plane

$$p: \mathbf{r} \cdot \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix} = 4$$



$$\text{shortest distance} = |\overrightarrow{AP} \cdot \hat{\mathbf{n}}|$$

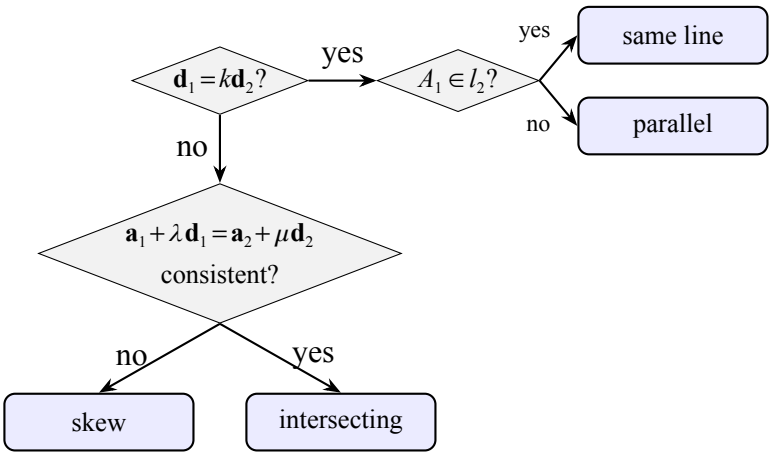
$$= \left| \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} \cdot \frac{1}{3} \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix} \right|$$

$$= \left| \frac{1}{3}(6+2-2) \right|$$

$$= 2$$

2.4 Relationships and Intersections

Line – Line $l_1: \mathbf{r}=\mathbf{a}_1+\lambda\mathbf{d}_1,\ l_2: \mathbf{r}=\mathbf{a}_2+\mu\mathbf{d}_2$



Example 2.7

$$l_1: \begin{pmatrix}1\\0\\2\end{pmatrix}+\lambda\begin{pmatrix}1\\2\\1\end{pmatrix},\ l_2: \begin{pmatrix}2\\12\\5\end{pmatrix}+\mu\begin{pmatrix}-1\\3\\0\end{pmatrix}$$

$$1+\lambda=2-\mu \quad —(1)$$

$$2\lambda=12+3\mu \quad —(2)$$

$$2+\lambda=5 \quad —(3)$$

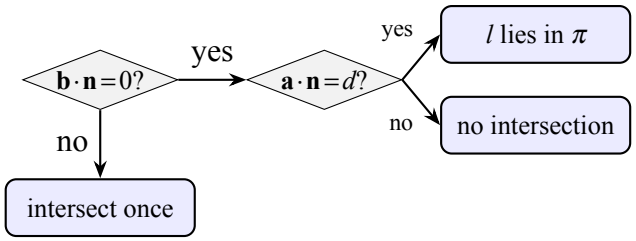
$(3):\ \lambda=3, \quad (1):\ \mu=-2$

$(2):\ 6=12-6$ ✓

$\therefore (4, 6, 5)$

Solve 2 of the 3 equations, sub into the 3rd: LHS = RHS ⇒ intersect; LHS ≠ RHS ⇒ skew.

Line – Plane $l: \mathbf{r}=\mathbf{a}+\lambda\mathbf{b},\ \pi: \mathbf{r}\cdot\mathbf{n}=d$



Example 2.8

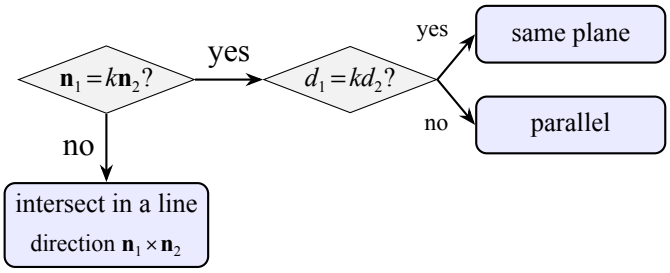
$$\mathbf{r}\cdot\begin{pmatrix}1\\2\\1\end{pmatrix}=5,\ \mathbf{r}=\begin{pmatrix}1\\3\\1\end{pmatrix}+\lambda\begin{pmatrix}1\\-2\\1\end{pmatrix}$$

$$\begin{pmatrix}1+\lambda\\3-2\lambda\\1+\lambda\end{pmatrix}\cdot\begin{pmatrix}1\\2\\1\end{pmatrix}=5$$

$$8-2\lambda=5 \Rightarrow \lambda=\tfrac{3}{2}$$

$\therefore (2.5, 0, 2.5)$

Plane – Plane $p_1: \mathbf{r}\cdot\mathbf{n}_1=d_1,\ p_2: \mathbf{r}\cdot\mathbf{n}_2=d_2$



Example 2.9

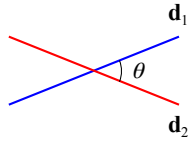
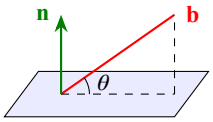
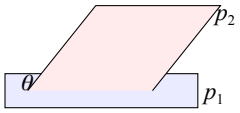
$3x+2y+z=10,\ x-2y+z=5$

GC: $y=\frac{5}{4}-\frac{1}{2}x, \quad z=\frac{15}{2}-2x$

$$\mathbf{r}=\begin{pmatrix}0\\1.25\\7.5\end{pmatrix}+\lambda\begin{pmatrix}2\\-1\\-4\end{pmatrix}$$

$\mathbf{d}=\mathbf{n}_1\times\mathbf{n}_2=(4,-2,-8)^{\mathrm{T}}$ // $(2,-1,-4)^{\mathrm{T}}$ checks the direction.

2.5 Angles

Line – Line	Line – Plane	Plane – Plane
		
$\cos\theta = \dfrac{\lvert\mathbf{d}_1\cdot\mathbf{d}_2\rvert}{\lvert\mathbf{d}_1\rvert\lvert\mathbf{d}_2\rvert}$	$\sin\theta = \left\lvert\dfrac{\mathbf{n}\cdot\mathbf{b}}{\lvert\mathbf{n}\rvert\lvert\mathbf{b}\rvert}\right\rvert$	$\cos\theta = \left\lvert\dfrac{\mathbf{n}_1\cdot\mathbf{n}_2}{\lvert\mathbf{n}_1\rvert\lvert\mathbf{n}_2\rvert}\right\rvert$
directions	sin: $\angle(\mathbf{n},\mathbf{b}) = 90^\circ - \theta$	normals

θ = acute angle.

Example 2.10

$3x-y+2z=3,\ y=2$

$$\cos\theta = \frac{\lvert -1\rvert}{\sqrt{14}\cdot 1}$$

$$\theta = 74.5^\circ$$

Example 2.11

$$\mathbf{r} = \begin{pmatrix}-2\\-4\\5\end{pmatrix} + \mu\begin{pmatrix}3\\5\\-4\end{pmatrix},\ \mathbf{r}\cdot\begin{pmatrix}0\\1\\0\end{pmatrix} = 1$$

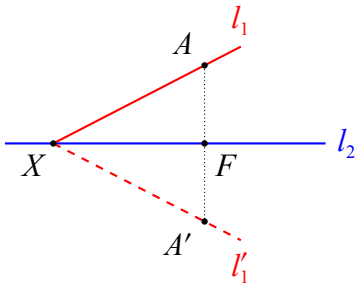
$$\sin\theta = \frac{5}{\sqrt{50}\cdot 1} = \frac{1}{\sqrt{2}}$$

$$\theta = 45^\circ$$

DRAFT: Your source lists 4.6 Angle Involving Lines, but that page is not in the PDF. The line–line formula above is the standard one; please confirm.

2.6 Reflection of a Line

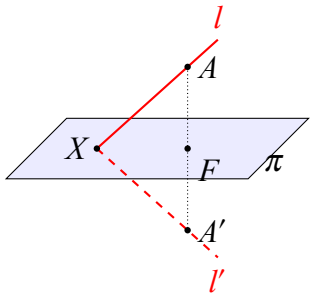
Line l_1 in line l_2 (intersect at X)



1. $X = l_1 \cap l_2$
2. $A \in l_1$: foot F on l_2
3. $\overrightarrow{OA'} = 2\overrightarrow{OF} - \overrightarrow{OA}$
4. l_1': $\mathbf{r} = \overrightarrow{OX} + \mu\overrightarrow{XA'}$

If $l \parallel \pi$: l' has the same direction, through A'.

Line l in plane π (intersect at X)



1. $X = l \cap \pi$
2. $A \in l$: foot F on π
3. $\overrightarrow{OA'} = 2\overrightarrow{OF} - \overrightarrow{OA}$
4. l': $\mathbf{r} = \overrightarrow{OX} + \mu\overrightarrow{XA'}$

Example 2.12

$l: \mathbf{r} = (7\mathbf{i}+3\mathbf{k}) + \lambda(2\mathbf{i}+\mathbf{k})$, $\pi: \mathbf{r}\cdot(\mathbf{i}+\mathbf{j}+2\mathbf{k}) = 1$

$$(7+2\lambda)+2(3+\lambda) = 1 \Rightarrow \lambda = -3$$

$$\therefore X = (1, 0, 0)$$

$$A(7, 0, 3): F = (5, -2, -1), \quad A' = (3, -4, -5) \quad (\S 2.3)$$

$$\overrightarrow{XA'} = \begin{pmatrix} 2 \\ -4 \\ -5 \end{pmatrix}$$

$$\therefore l': \mathbf{r} = \mathbf{i} + \mu(2\mathbf{i} - 4\mathbf{j} - 5\mathbf{k}), \ \mu \in \mathbb{R}$$

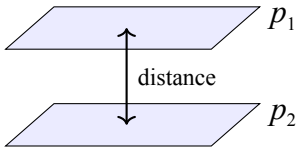
DRAFT: 4.7 Reflection (Line about Line) is listed in your source but its page is missing; the steps on the left are the standard method. Add an example from an official answer sheet?

2.7 Distance Between Parallel Planes

$$p_1: \mathbf{r}\cdot\mathbf{n} = d_1, \qquad p_2: \mathbf{r}\cdot\mathbf{n} = d_2$$

Distance

$$\text{distance} = \left| \frac{d_1}{|\mathbf{n}|} - \frac{d_2}{|\mathbf{n}|} \right| \qquad \text{plane to } O = \frac{|k|}{|\mathbf{n}|}$$



Same $\mathbf{n}$ for both planes first. $\mathbf{r}\cdot\mathbf{n} = 0$: plane through O.

Example 2.13

$x+2y-3z=2$, $2x+4y-6z=-10$

$$2x+4y-6z=-10 \Rightarrow x+2y-3z=-5$$

$$\text{distance} = \left| \frac{2}{\sqrt{14}} - \frac{-5}{\sqrt{14}} \right| = \frac{7}{\sqrt{14}}$$